Introduction

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Adapted from slides of CS224N: Natural Language Processing with Deep Learning, Stanford

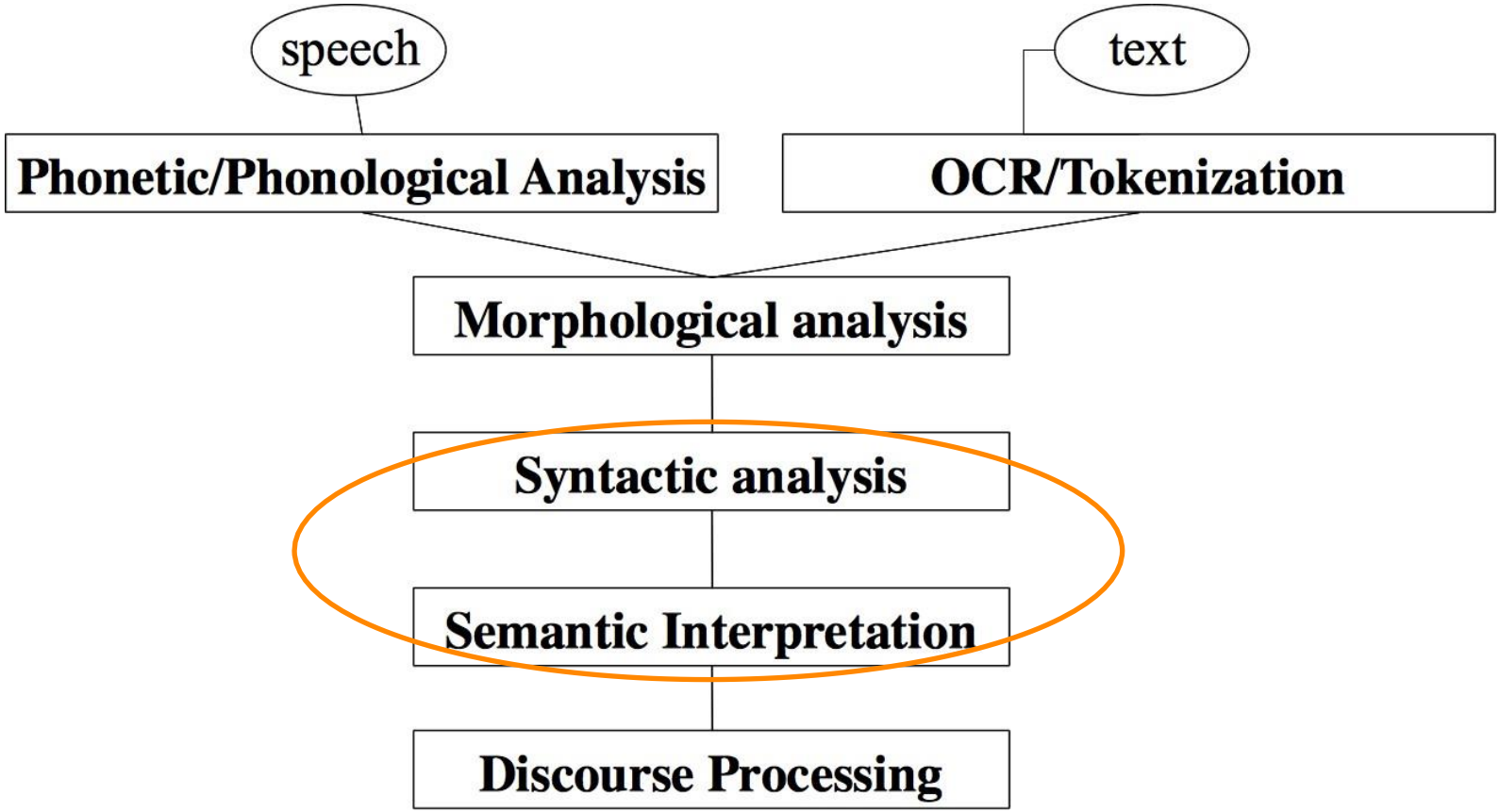
Expected lesson learning outcomes

LLO1. Hiểu được các đặc trưng cơ bản của ngôn ngữ tự nhiên, các bài toán điển hình trong NLP
LLO2. Nêu được một số ứng dụng NLP
LLO3. Giải thích được các ưu điểm của học sâu trong giải quyết các bài toán NLP

1. What is Natural Language Processing (NLP)?

- **Natural language processing** is a field at the intersection of
  - computer science
  - artificial intelligence
  - and linguistics.
- **Goal:** for computers to process or "understand" natural language in order to perform tasks that are useful, e.g.,
  - Performing Tasks, like making appointments, buying things
  - Language translation
  - Question Answering
    - Siri, Google Assistant, Facebook M, Cortana …
- Fully **understanding and representing** the **meaning** of language (or even defining it) is a difficult goal.
- Perfect language understanding is AI-complete

NLP Levels


speech
text
Phonetic/Phonological Analysis
OCR/Tokenization
Morphological analysis
Syntactic analysis
Semantic Interpretation
Discourse Processing

(A tiny sample of) NLP Applications

Applications range from simple to complex:

- Spell checking, keyword search, finding synonyms

- Extracting information from websites such as
  - product price, dates, location, people or company names
- Classifying: reading level of school texts, positive/negative sentiment of longer documents

- Machine translation
- Spoken dialog systems
- Complex question answering

NLP in industry … is taking off

- Search (written and spoken)
- Online advertisement matching
- Automated/assisted translation
- Sentiment analysis for marketing or finance/trading
- Speech recognition
- Chatbots / Dialog agents
  - Automating customer support
  - Controlling devices
  - Ordering goods

What's special about human language?

A human language is a system **specifically constructed to convey the speaker/writer's meaning**

- Not just an environmental signal, it's a deliberate communication
- Using an encoding which little kids can quickly learn (**amazingly!**) and which changes

A human language is mostly a **discrete/symbolic/categorical signaling system**

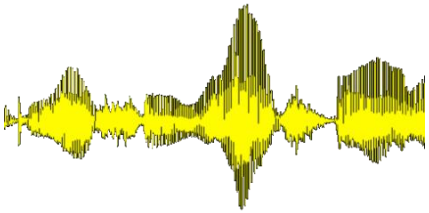
- rocket =  ; violin = 
- Presumably because of greater signaling reliability
- Symbols are not just an invention of logic / classical AI!

What's special about human language?

The categorical symbols of a language can be encoded as a signal for communication in several ways:

- Sound
- Gesture
- Writing/Images

The symbol is invariant across different encodings!





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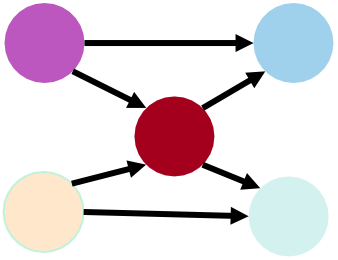
What's special about human language?

A human language is a **symbolic/categorical signaling system**

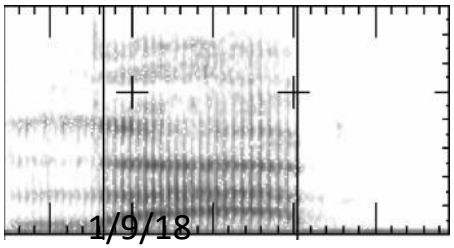
However, a brain encoding appears to be a **continuous pattern of activation**, and the symbols are transmitted via **continuous signals** of sound/vision

The large vocabulary, symbolic encoding of words creates a problem for machine learning – sparsity!

We will explore a continuous encoding pattern of thought

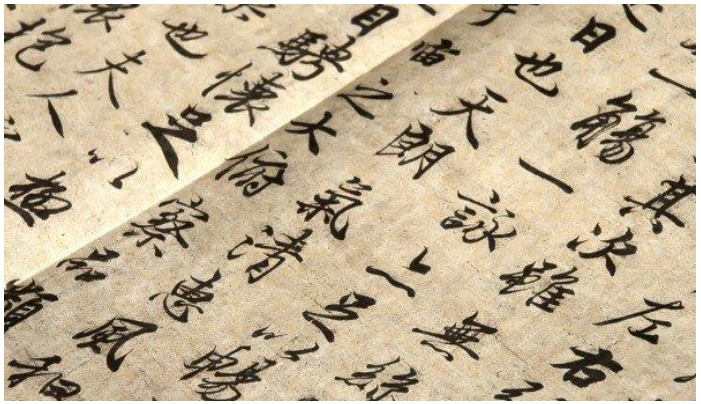


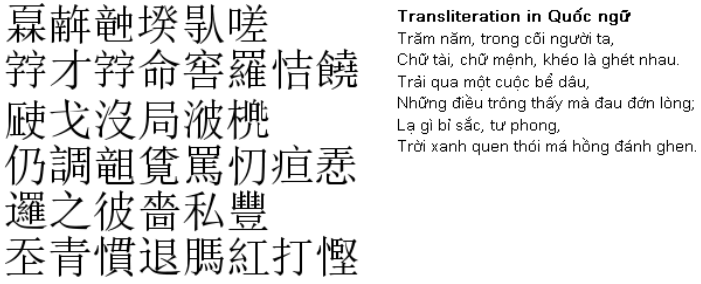
lab



Lịch sử chữ viết tiếng Việt

- Chữ Hán
  - Còn gọi là chữ Nho, Hán tự
  - Dùng chính thức trong các triều đại phong kiến ở Việt Nam cho đến giữa thế kỷ 19
- Chữ Nôm
  - Dựa trên chữ Hán để phiên âm các từ tiếng Việt bản địa
  - Sử dụng chủ yếu trong các tác phẩm văn học





Transliteration in Quốc ngữ

Trăm năm, trong cõi người ta,
Chữ tài, chữ mệnh, khéo là ghét nhau.
Trải qua một cuộc bể dâu,
Những điều trông thấy mà đau đớn lòng;
Lạ gì bỉ sắc, tư phong,
Trời xanh quen thói má hồng đánh ghen.

Chữ quốc ngữ

- Được các nhà truyền giáo người Bồ Đào Nha phát triển vào thế kỷ 17
- Trong khoảng 200 năm đầu chủ yếu được sử dụng trong cộng đồng Công giáo
- Cuối thế kỷ 19, đầu thế kỷ 20 bảng chữ cái được sửa đổi thêm và dần trở thành một phần của giáo dục phổ thông



Các ngôn ngữ ở Việt Nam

- Số lượng các ngôn ngữ ở ví dụ: 54. Đây là con số chính thức của nhà nước. Trên thực tế có khoảng 100 ngôn ngữ, phương ngữ.
- Các họ ngôn ngữ ở Việt Nam:

  Nam Á: tiếng Việt, tiếng Mường, tiếng Khmer, tiếng Khơ mú, tiếng Katu, tiếng Bana,…

  Nam Đảo: tiếng Chăm/ Chàm, tiếng Ê đê, tiếng Raglai, tiếng Chu ru, tiếng In đô, tiếng Ma lai,…

  Thái – Kađai: tiếng Tày, tiếng Nùng, tiếng Thái, tiếng Lào, Lự, tiếng Choang,…

  Mông – Dao: tiếng Hmông, tiếng Dao, tiếng Pà thẻn,…

  Hán – Tạng: tiếng Hán, tiếng Tạng, tiếng Lô lô,…



Các ngôn ngữ ở Việt Nam

Các ngôn ngữ đã có chữ viết: Tày, Thái, Hoa Khmer, Bana, Hmông, Chăm, Mnông, Mường (bộ gõ do nhóm UET làm hiện đang dùng cho Báo điện tử Hòa Bình),…

Chữ viết cổ: Khmer, Chăm, Hoa, Thái, Lào,...

Chữ viết Latin: Tày, Thái, Banan, Ê đê, Cơ ho, Mnông, Cơ tu,….

Số lượng ngôn ngữ có chữ viết: khoảng 32/54

Các ngôn ngữ chưa có chữ viết: Khơ mú, Kháng, Pà thẻn, Bố ý, Cống, Si la, Xinh mun,…

Các ngôn ngữ đã được đưa vào trong nhà trường: HMông, Chăm, Khơ Me, Jrai, Ba Na, Ê Đê, Thái, Mnông…



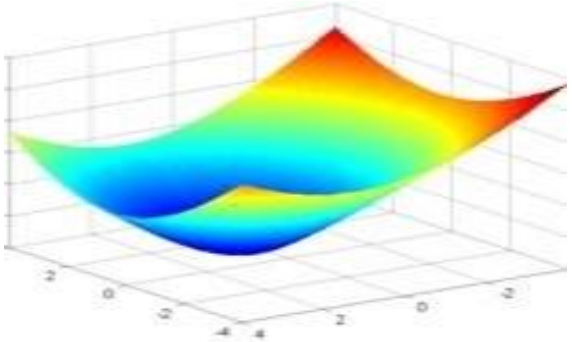
2. What's Deep Learning (DL)?

- **Deep learning** is a subfield of **machine learning**

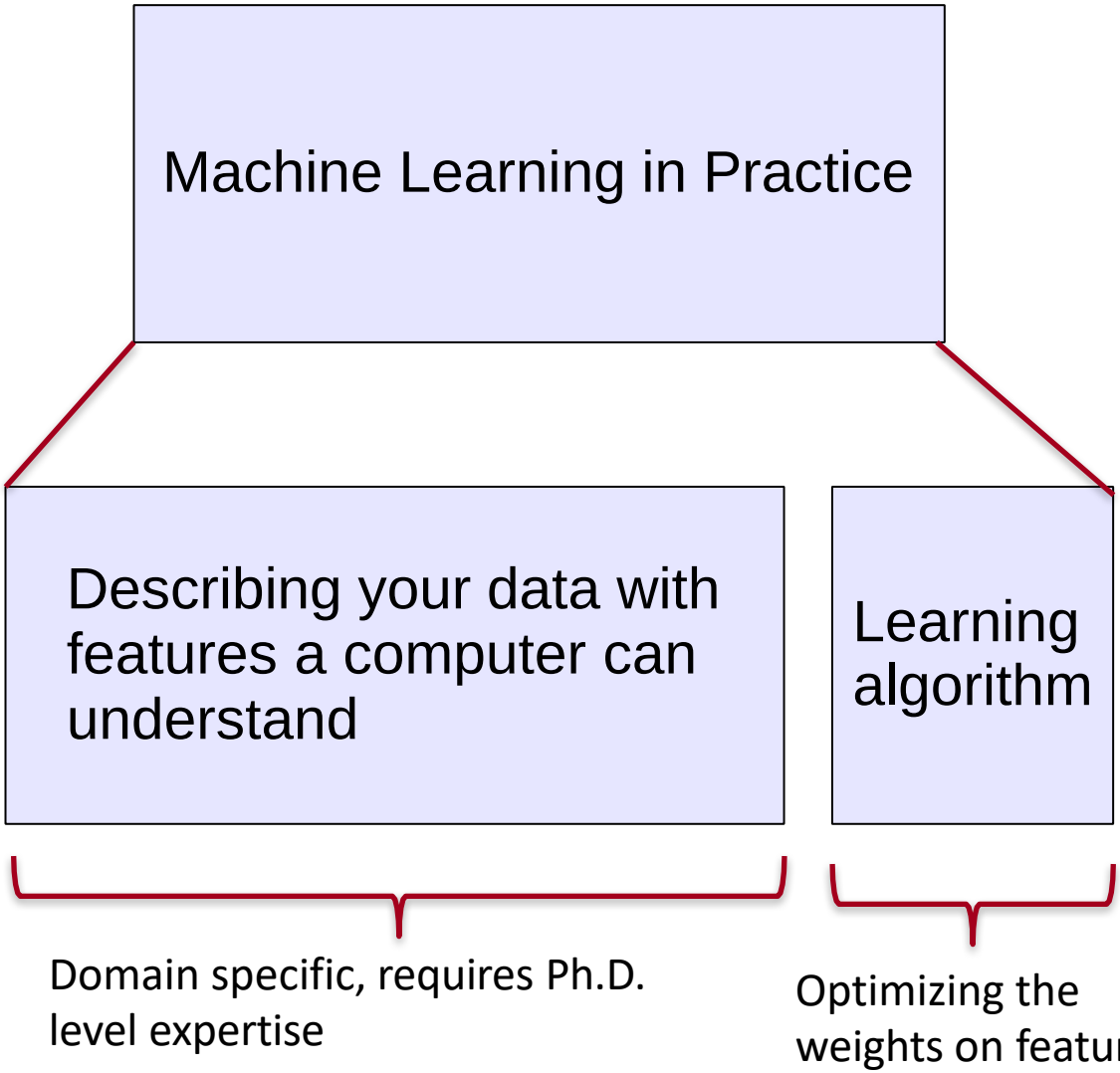
- Most machine learning methods work well because of **human-designed representations** and **input features**
  - For example: features for finding named entities like locations or organization names (Finkel et al., 2010):

Feature	NER
Current Word	C
Previous Word	C
Next Word	C
Current Word Character n-gram	all
Current POS Tag	C
Surrounding POS Tag Sequence	C
Current Word Shape	C
Surrounding Word Shape Sequence	C
Presence of Word in Left Window	size 4
Presence of Word in Right Window	size 4

- Machine learning becomes just optimizing weights to best make a final prediction

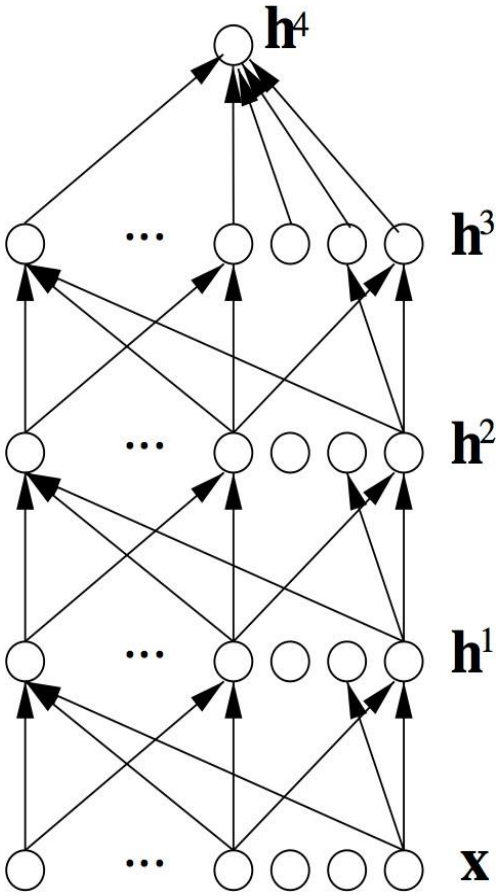


Machine Learning vs. Deep Learning



What's Deep Learning (DL)?

- In contrast to standard machine learning,
- Representation learning attempts to automatically learn good features or representations
- Deep learning algorithms attempt to learn (multiple levels of) representations (here: h^1,h^2,h^3) and an output (h^4)
- From "raw" inputs **x** (e.g. sound, pixels, characters, or words)



On the history of "Deep Learning"

- We will focus on different kinds of **neural networks**
- The dominant model family inside deep learning

- Only clever terminology for stacked logistic regression units?
  - Maybe, but interesting modeling principles (end-to-end) and actual connections to neuroscience in some cases.
  - Recently: Differentiable Programming – becomes clear later

- We will not take a historical approach but instead focus on methods which work well on NLP problems now
- For a long history of deep learning models (starting ~1960s), see: Deep Learning in Neural Networks: An Overview by Jürgen Schmidhuber

Reasons for Exploring Deep Learning

- Manually designed features are often over-specified, incomplete and take a long time to design and validate
- **Learned Features** are easy to adapt, fast to learn
- Deep learning provides a very flexible, (almost?) universal, learnable framework for **representing** world, visual and linguistic information.
- Deep learning can learn **unsupervised** (from raw text) and **supervised** (with specific labels like positive/negative)

Reasons for Exploring Deep Learning

- In ~2010 **deep** learning techniques started outperforming other machine learning techniques. Why this decade?

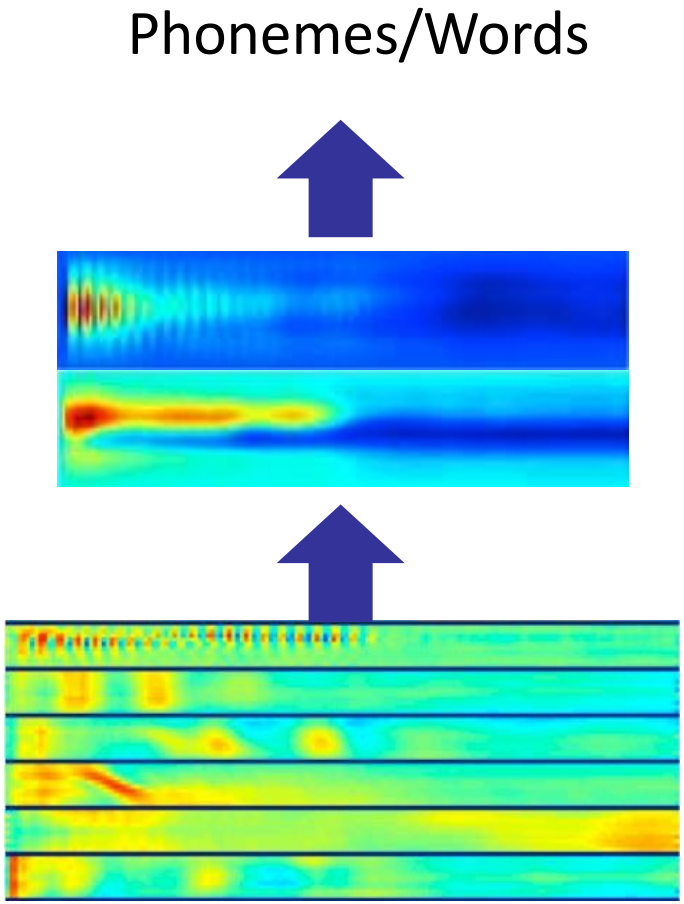
- Large amounts of training data favor deep learning
- Faster machines and multicore CPU/GPUs favor Deep Learning
- New models, algorithms, ideas
  - Better, more flexible learning of intermediate representations
  - Effective end-to-end joint system learning
  - Effective learning methods for using contexts and transferring between tasks
  - Better regularization and optimization methods

→ **Improved performance** (first in speech and vision, then NLP)

Deep Learning for Speech

- The first breakthrough results of "deep learning" on large datasets happened in speech recognition
- Context-Dependent Pre-trained Deep Neural Networks for Large Vocabulary Speech Recognition Dahl et al. (2010)

Acoustic model and WER	RT03S FSH	Hub5 SWB
Traditional features	**27.4**	**23.6**
Deep Learning	**18.5** (–33%)	**16.1** (–32%)

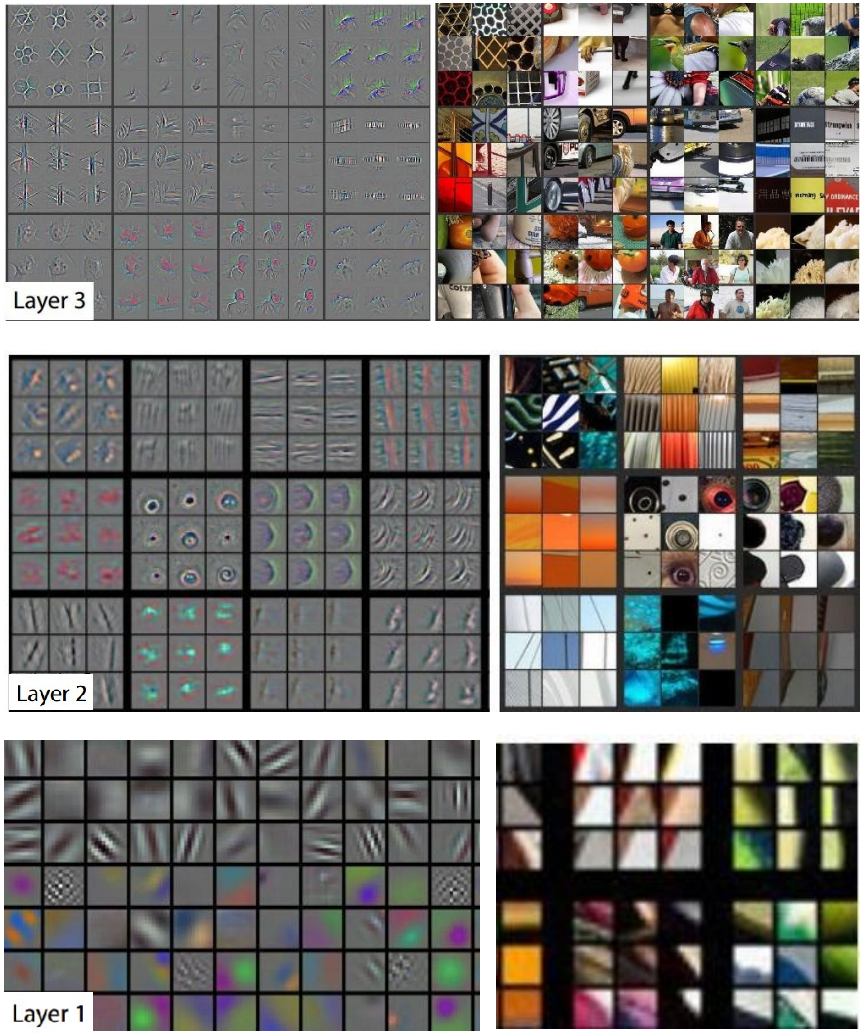


Deep Learning for Computer Vision

First major focus of deep learning groups was computer vision

The breakthrough DL paper: ImageNet Classification with Deep Convolutional Neural Networks by Krizhevsky, Sutskever, & Hinton, 2012, U. Toronto. 37% error red.





Zeiler and Fergus (2013)



What do we hope to teach?

1. An understanding of and ability to use the effective modern methods for deep learning
   - Basics first, then key methods used in NLP: Recurrent networks, attention, etc.
2. Some big picture understanding of human languages and the difficulties in understanding and producing them
3. An understanding of and ability to build systems (in TensorFlow) for some of the major problems in NLP:
   - Word similarities, parsing, machine translation, entity recognition, question answering, sentence comprehension

4. Why is NLP hard?

- Complexity in representing, learning and using linguistic/situational/contextual/world/visual knowledge
- But interpretation depends on these

- Human languages are ambiguous (unlike programming and other formal languages)

- E.g. “Ông già đi nhanh quá.”
  “Đêm hôm qua cầu gãy.”



Why NLP is difficult: Real newspaper headlines/tweets

1. “Boy paralyzed after tumor fights back to gain black belt"

2. "Enraged cow injures farmer with axe"

3. "Juvenile Court to Try Shooting Defendant"

5. Deep NLP = Deep Learning + NLP

Combine ideas and goals of NLP with using representation learning and deep learning methods to solve them

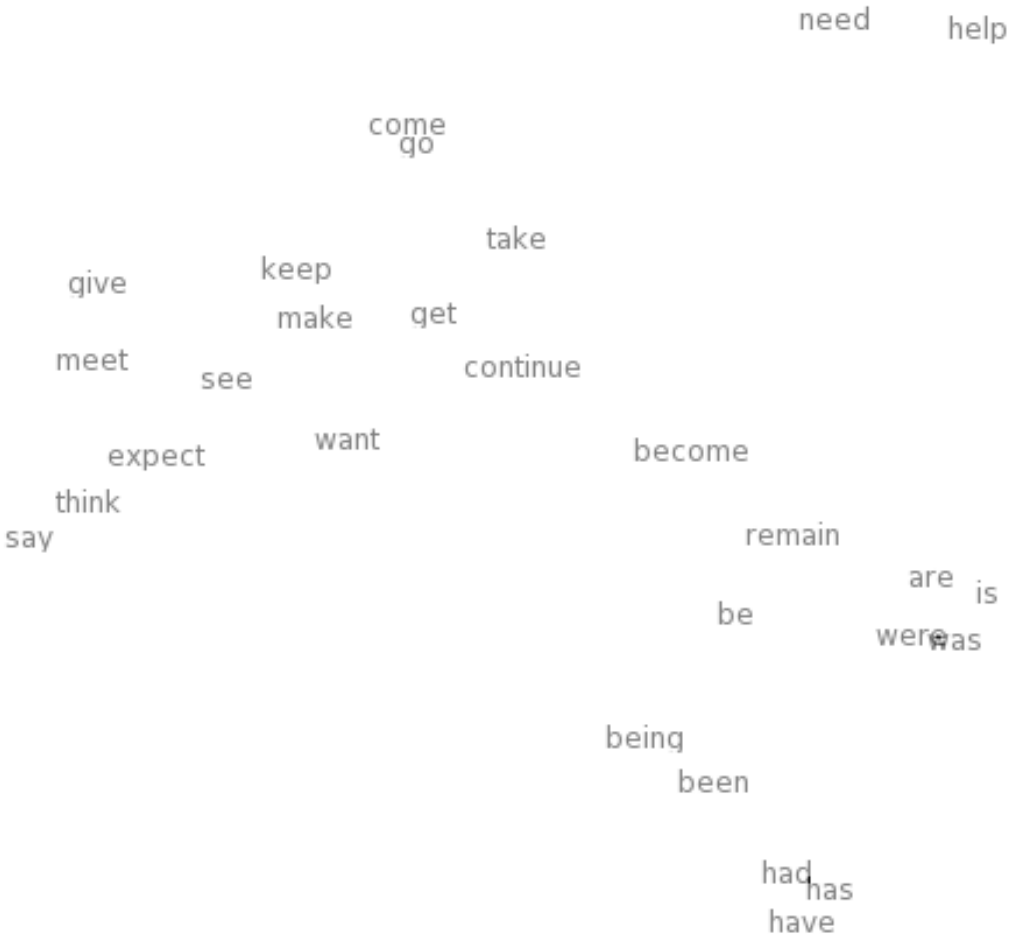
Several big improvements in recent years in NLP

- **Linguistic levels**: (speech), words, syntax, semantics
- **Intermediate tasks/tools:** parts-of-speech, entities, parsing
- **Full applications**: sentiment analysis, question answering, dialogue agents, machine translation

Word meaning as a neural word vector – visualization

expect =

$$\begin{pmatrix} 0.286 \\ 0.792 \\ -0.177 \\ -0.107 \\ 0.109 \\ -0.542 \\ 0.349 \\ 0.271 \\ 0.487 \end{pmatrix}$$



Word similarities

Nearest words to frog:

1. frogs
2. toad
3. litoria
4. leptodactylidae
5. rana
6. lizard
7. eleutherodactylus



litoria



leptodactylidae



rana



eleutherodactylus

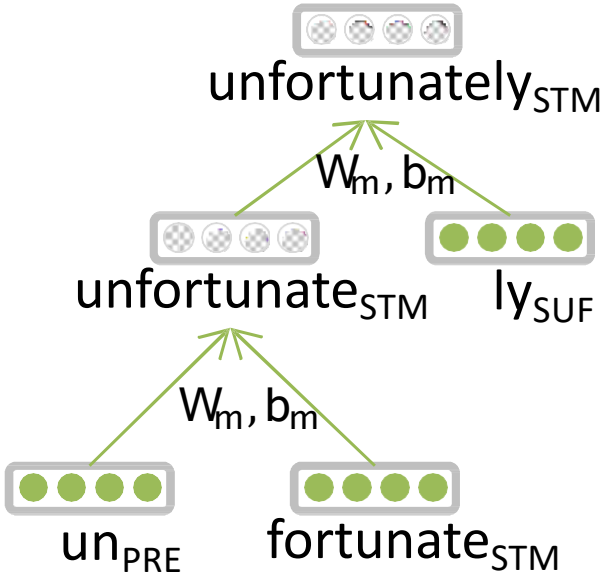
http://nlp.stanford.edu/projects/glove/

Representations of NLP Levels: Morphology

- Traditional: Words are made of morphemes

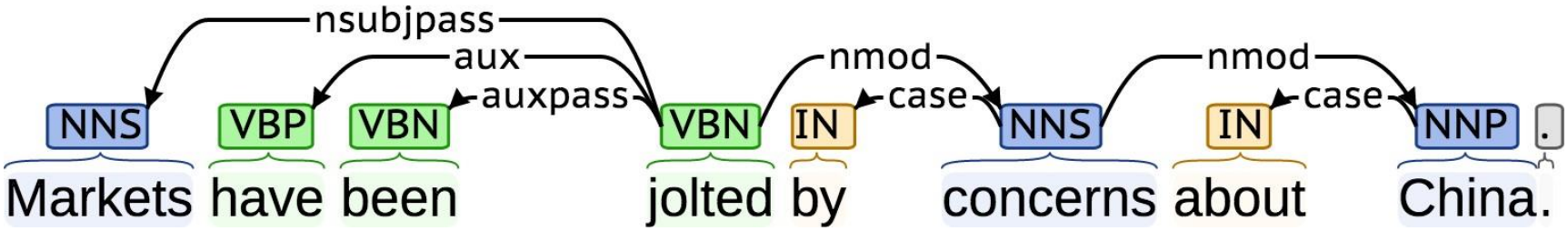
prefix	stem	suffix
un	interest	ed

- DL:
  - every morpheme is a vector
  - a neural network combines two vectors into one vector
  - Luong et al. 2013



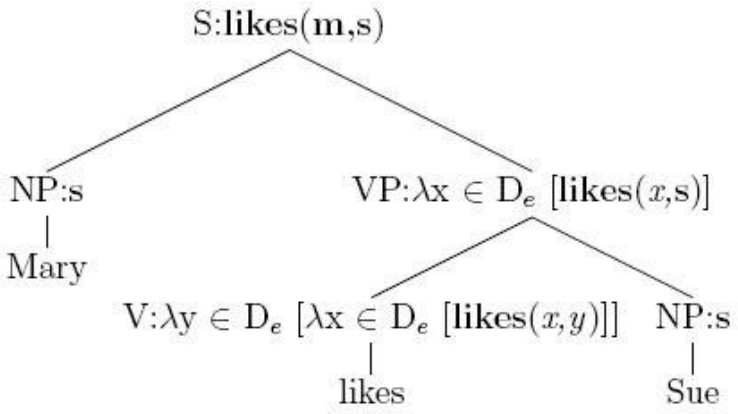
NLP Tools: Parsing for sentence structure

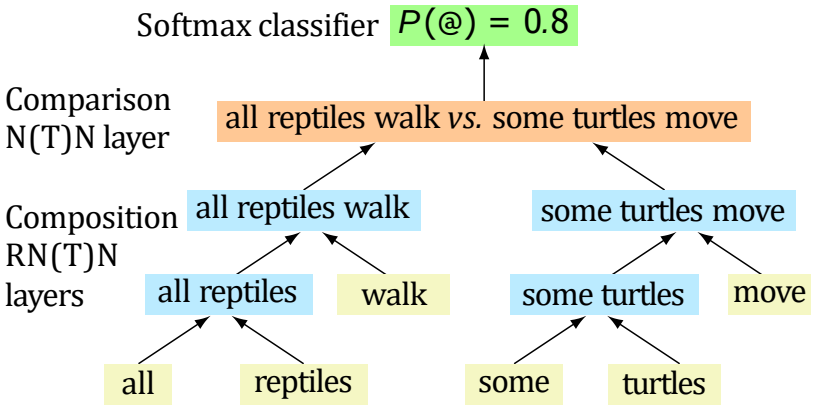
- Neural networks can accurately determine the grammatical structure of sentences
- This supports interpretation and may help in disambiguation



Representations of NLP Levels: Semantics

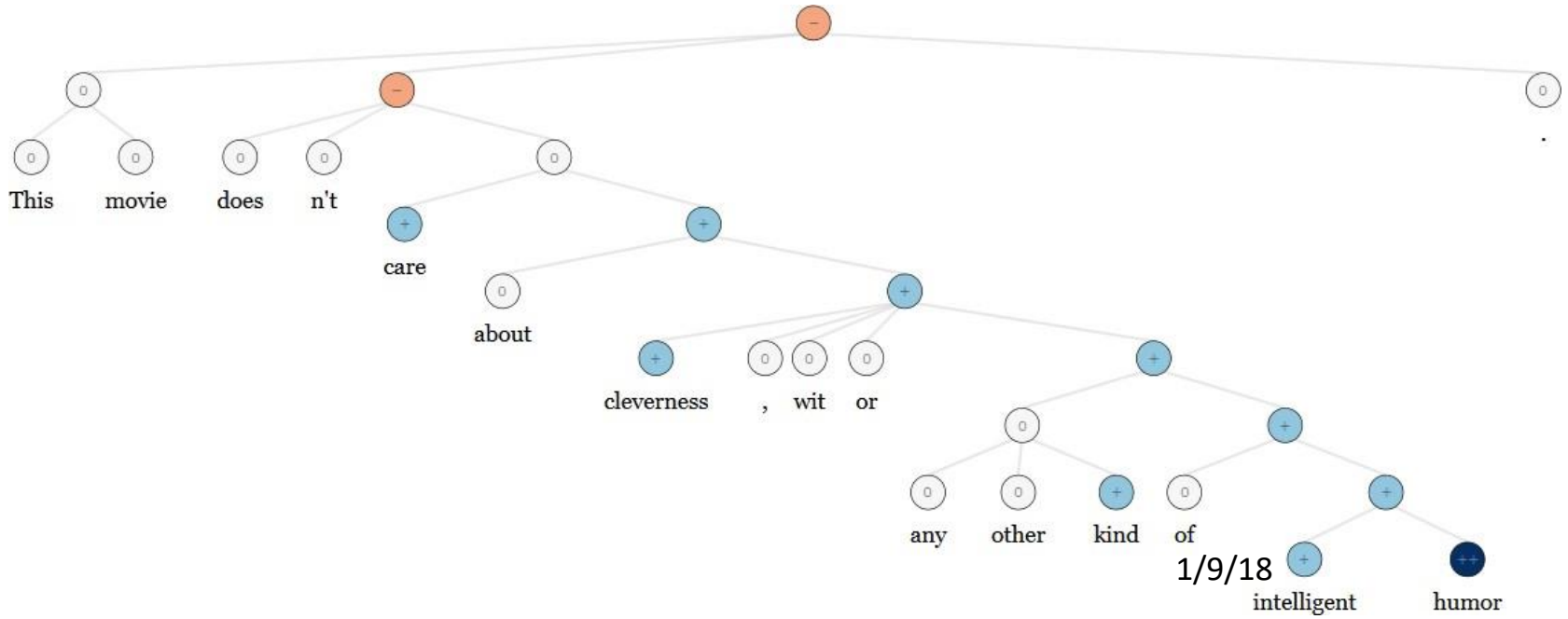
- Traditional: Lambda calculus
  - Carefully engineered functions
  - Take as inputs specific other functions
  - No notion of similarity or fuzziness of language
- DL:
  - Every word and every phrase and every logical expression is a vector
  - a neural network combines two vectors into one vector
  - Bowman et al. 2014





NLP Applications: Sentiment Analysis

- Traditional: Treat sentence as a bag-of-words (ignore word order); consult a curated list of "positive" and "negative" words to determine sentiment of sentence. Need hand-designed features to capture negation! --> Ain’t gonna capture everything 😖
- Same deep learning model that could be used for morphology, syntax and logical semantics → RecursiveNN (aka TreeRNNs)



Question Answering

- Traditional: A lot of feature engineering to capture world and other knowledge, e.g., regular expressions, Berant et al. (2014)

Is main verb trigger?

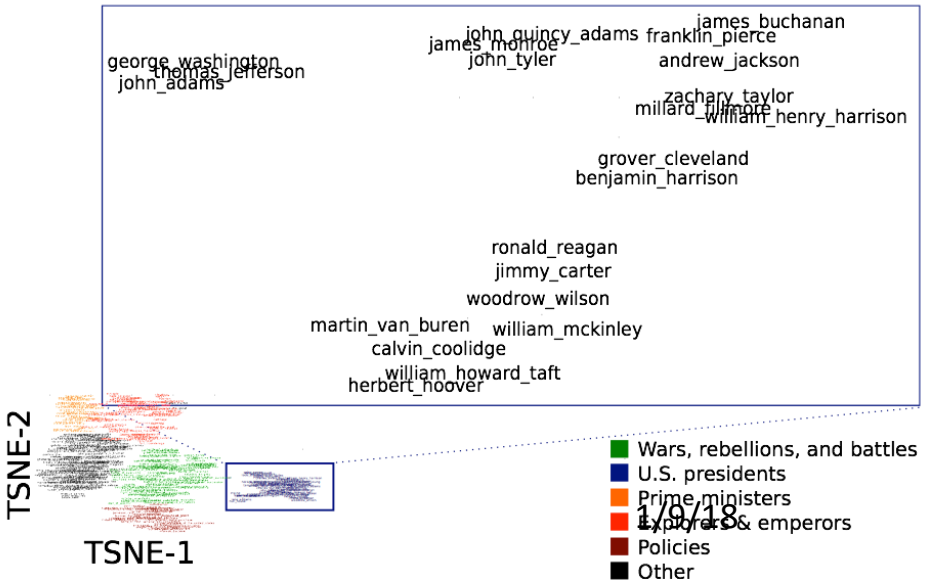
Yes

Condition	Regular Exp.
Wh- word subjective?	AGENT
Wh- word object?	THEME

No

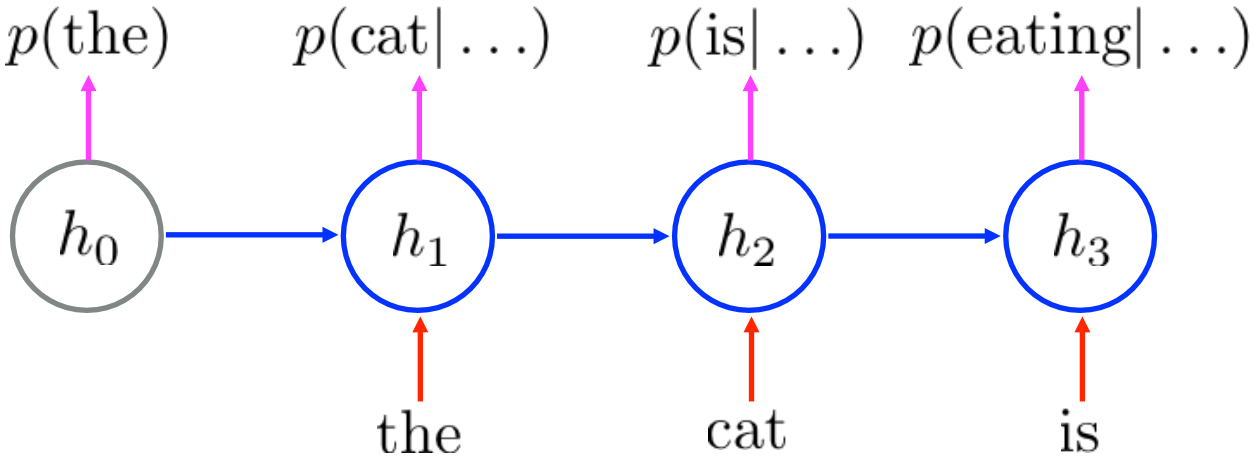
Condition	Regular Exp.
default	(ENABLE|SUPER)$^+$
DIRECT	(ENABLE|SUPER)
PREVENT	(ENABLE|SUPER)* PREVENT(ENABLE|SUPER)*

- DL: Again, a deep learning architecture can be used!
- Facts are stored in vectors



Dialogue agents / Response Generation

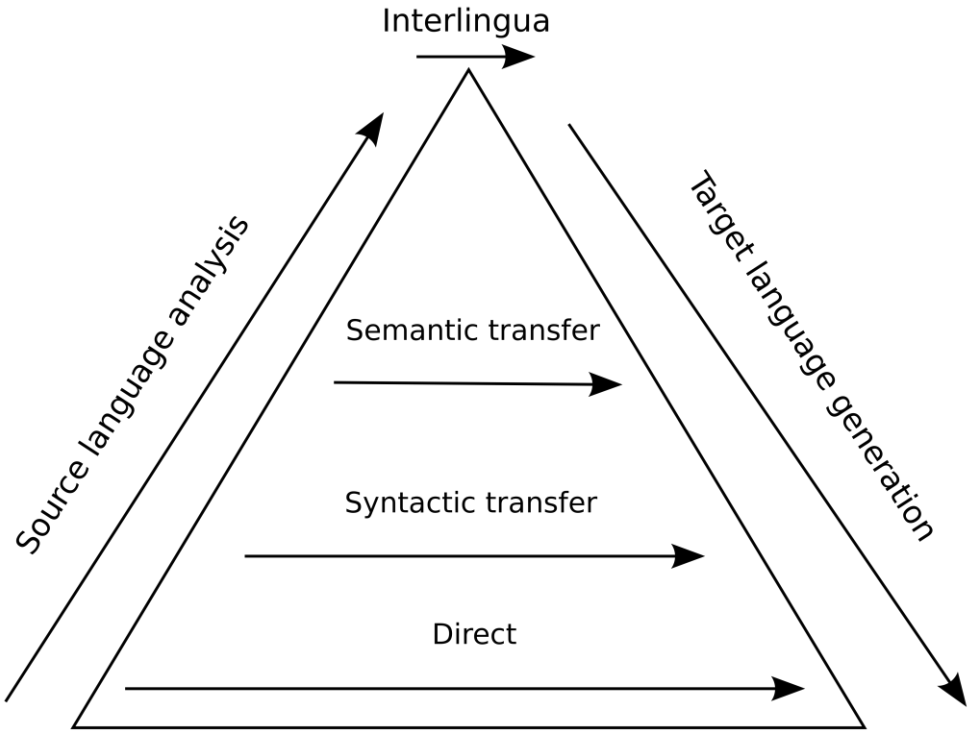
- A simple, successful example is the auto-replies available in the Google Inbox app
- An application of the powerful, general technique of **Neural Language Models**, which are an instance of Recurrent Neural Networks



Machine Translation

- Many levels of translation have been tried in the past:

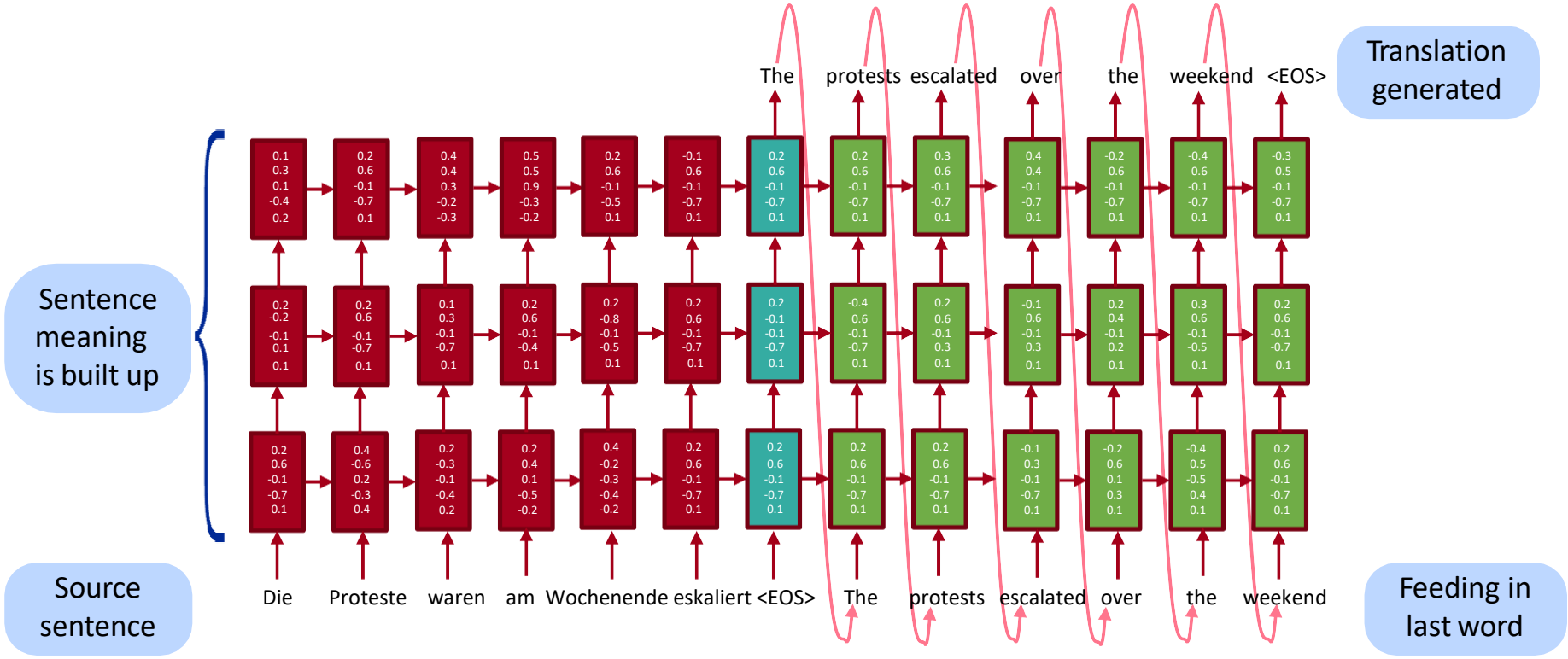
- Traditional MT systems are very large complex systems



- What do you think is the interlingua for the DL approach to translation?

Neural Machine Translation

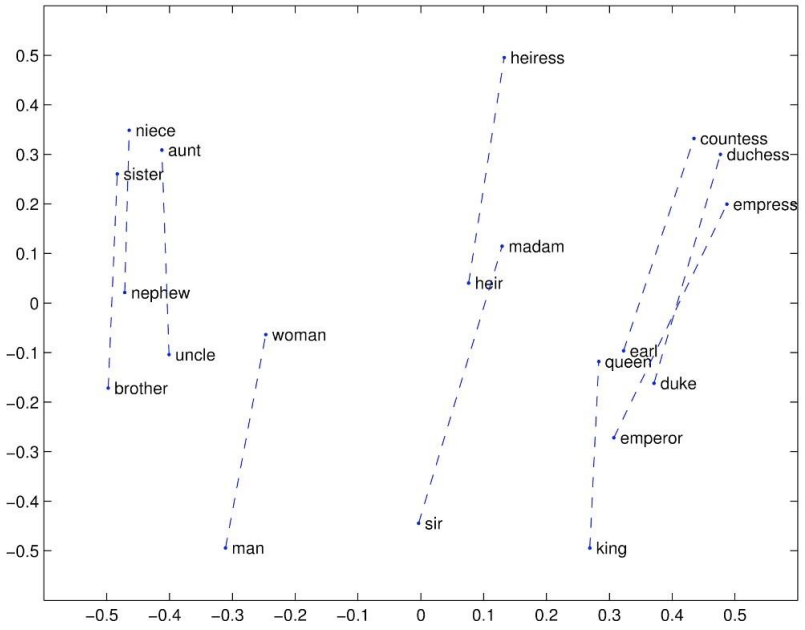
Source sentence is mapped to **vector**, then output sentence generated
[Sutskever et al. 2014, Bahdanau et al. 2014, Luong and Manning 2016]



Now live for some languages in Google Translate (etc.), with big error reductions!

Conclusion: Representation for all levels? Vectors

We will study in the next lecture how we can learn vector representations for words and what they actually **represent.**



Next week: how neural networks work and how they can use these vectors for all NLP levels and many different applications